

CAPE COLUMBINE, South Africa

WMO: 687120

Lat: 32.8278S

Lon: 17.8558E

Elev: 180

StdP: 14.60

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	46.9	48.3	33.2	27.9	62.5	36.7	32.1	59.8	29.6	54.2	25.6	54.3	10.4	150	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.1	80.1	62.5	76.6	61.1	73.3	60.9	66.3	73.1	65.2	70.9	64.2	69.1	11.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	90.4	68.0	63.2	87.2	66.9	62.3	84.6	66.0	31.0	73.6	30.2	71.0	29.5	69.2	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.7	23.6	21.8	43.9	88.9	2.1	2.3	42.4	90.5	41.1	91.9	39.9	93.2	38.4	94.9		
(5)			41.0	68.6	2.1	1.3	39.5	69.5	38.2	70.2	37.1	71.0	35.6	71.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.9	63.3	63.6	62.4	61.4	58.7	56.8	55.9	55.8	57.8	59.5	60.7	62.7	(6)
(7)		DBStd	4.50	2.62	3.08	3.83	5.40	3.66	3.55	3.82	3.22	3.76	3.14	2.76	2.99	(7)
(8)		HDD50	2	0	0	0	0	1	0	1	0	0	0	0	0	(8)
(9)		HDD65	2003	67	58	100	140	201	250	282	286	225	177	134	83	(9)
(10)		CDD50	3605	412	382	386	343	271	203	184	181	233	294	321	395	(10)
(11)		CDD65	131	15	20	21	33	6	3	1	1	8	6	4	13	(11)
(12)		CDH74	610	28	72	129	181	26	11	4	10	59	31	18	41	(12)
(13)		CDH80	119	4	9	29	40	1	0	0	0	18	8	4	6	(13)
(14)	Wind	WSAvg	12.1	13.8	13.0	12.1	11.4	10.2	10.7	10.8	11.1	12.0	12.8	14.0	13.6	(14)
(15)	Precipitation	PrecAvg	10.2	0.2	0.2	0.3	0.8	1.3	2.0	1.8	1.5	0.9	0.4	0.4	0.3	(15)
(16)		PrecMax	14.8	1.5	1.0	1.0	2.8	3.3	6.9	4.4	3.3	3.0	1.9	1.6	0.9	(16)
(17)		PrecMin	4.1	0.0	0.0	0.0	0.1	0.1	0.3	0.3	0.4	0.2	0.0	0.0	0.0	(17)
(18)		PrecStd	2.4	0.3	0.3	0.3	0.7	0.8	1.3	0.9	0.7	0.7	0.5	0.4	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.1	83.2	84.2	85.1	77.8	74.8	72.7	74.5	81.4	79.3	75.6	79.2	(19)
(20)			MCWB	64.1	65.5	65.7	63.8	57.8	57.7	55.9	58.2	58.9	60.6	60.6	62.6	(20)
(21)		2%	DB	72.4	76.4	77.4	78.8	71.4	68.7	68.4	66.7	70.8	72.0	70.6	72.7	(21)
(22)			MCWB	64.3	63.1	63.4	60.7	56.4	55.6	55.1	54.4	55.8	58.0	59.8	63.4	(22)
(23)		5%	DB	70.0	72.0	71.8	74.4	67.8	64.5	65.3	62.9	64.7	66.8	67.5	70.0	(23)
(24)			MCWB	63.5	63.7	63.0	59.8	56.2	55.0	54.2	54.2	55.9	58.2	59.9	63.0	(24)
(25)		10%	DB	68.2	69.4	68.0	69.4	63.8	61.2	61.9	60.2	62.0	64.5	65.7	68.2	(25)
(26)			MCWB	62.7	63.2	62.3	58.4	56.4	54.8	54.2	55.1	56.3	58.2	59.6	62.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.3	67.8	67.5	65.6	61.5	60.1	59.4	59.8	60.8	62.8	63.9	66.6	(27)
(28)			MCDWB	73.1	74.9	77.2	80.5	67.5	65.8	64.7	65.7	71.0	72.9	69.3	72.4	(28)
(29)		2%	WB	65.7	66.2	65.5	63.4	60.1	58.9	57.9	58.0	59.1	61.0	62.4	65.1	(29)
(30)			MCDWB	70.1	72.7	72.5	73.2	64.4	62.9	61.8	61.6	64.3	67.0	67.1	70.3	(30)
(31)		5%	WB	64.5	64.8	64.0	61.9	59.2	57.7	56.9	56.8	58.0	59.8	61.3	63.9	(31)
(32)			MCDWB	68.3	69.6	69.6	69.1	62.5	60.8	60.5	59.8	62.2	64.6	65.8	68.5	(32)
(33)		10%	WB	63.6	63.7	62.9	60.7	58.4	56.7	56.0	55.9	57.2	58.8	60.3	62.9	(33)
(34)			MCDWB	67.1	67.8	67.3	66.1	61.3	59.5	59.2	58.7	60.6	62.8	64.5	67.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	10.1	9.4	9.4	8.2	8.3	8.9	8.7	9.2	9.6	9.7	10.0	(35)
(36)			MCDBR	12.3	14.6	14.5	17.2	16.1	13.3	15.2	14.3	14.3	13.4	12.2	12.4	(36)
(37)			MCWBR	6.1	6.7	5.9	6.7	6.5	7.4	8.1	8.0	7.1	6.5	6.1	6.1	(37)
(38)		5% WB	MCDWB	10.7	11.8	11.6	12.7	9.6	8.0	9.5	9.4	10.1	10.4	10.7	10.8	(38)
(39)			MCWBR	6.1	6.4	6.1	6.3	5.9	5.6	6.4	6.3	6.3	5.9	6.2	5.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.335	0.336	0.329	0.317	0.315	0.308	0.311	0.327	0.336	0.346	0.331	0.335	(40)	
(41)		taud	2.564	2.571	2.584	2.610	2.607	2.592	2.581	2.539	2.472	2.463	2.551	2.554	(41)	
(42)		Ebn at Noon	318	312	303	290	274	267	272	282	295	305	318	319	(42)	
(43)		Edh at Noon	34	33	30	26	23	22	24	28	33	36	34	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2642	2373	1926	1416	994	856	952	1191	1625	2097	2482	2664	(44)	
(45)		RadStd	71	66	68	75	65	48	55	52	61	89	70	86	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110

Nomenclature: See separate page